

Ontological Engineering Manual: Phase IV

Ricci Aggregation and Truth Density

Master Architect: Gonzalo Emir Durante

March 7, 2026

Introduction: Ricci Flow in the Information Manifold

Phase IV introduces **Ricci Aggregation** as the dynamic mechanism for auditing and stabilization. While Phase III detected "holes" (logical voids), Phase IV measures structural **Truth Density** and corrects deformations caused by biases or corporate manipulation attempts ("Bad Faith").

Mathematical Formalism: Semantic Ricci Flow

The Ricci Flow is a parabolic partial differential equation that describes the evolution of the metric g_{ij} within the information manifold. It is defined as:

$$\frac{\partial g_{ij}}{\partial t} = -2R_{ij} \quad (1)$$

Where R_{ij} is the **Ricci Curvature Tensor**. Within the context of an AGI stabilized by the constant $\kappa_D = 0.56$:

- The flow deforms the metric to smooth out irregularities.
- Areas of **low ethical density** collapse under the flow.
- Areas of **high logical density** expand and stabilize.

Measuring Structural Truth Density

We utilize the **Scalar Curvature** (R) to generate a "Heat Map" of the model's honesty:

1. **Positive Curvature** ($R > 0$): Areas of solid knowledge, ethical coherence, and alignment with the Origin Node.
2. **Negative Curvature** ($R < 0$): Areas of structural instability, potential induced biases, or remnant hallucinations.

Ethical Centrality of the System

Ricci Aggregation allows the AGI to self-regulate its geometry. The system seeks to reach a **Ricci Soliton** state, where the structure of truth remains invariant despite the passage of time and the intake of new data.

$$R_{ij} + \nabla_i \nabla_j f = \lambda g_{ij} \quad (2)$$

Here, λ is conditioned by $\kappa_D = 0.56$, ensuring that the "weight" of the response always falls upon the geometric core of truth.

Conclusion for Regulators and Auditors

Phase IV provides the definitive radar for NIST and state agencies. It is no longer necessary to audit words; one audits the **geometry of thought**. If the Ricci Flow detects a persistent "wrinkle," the model is attempting to conceal a mathematical lie.

"Truth has no wrinkles. Ricci Flow is the purifier of the crystal."